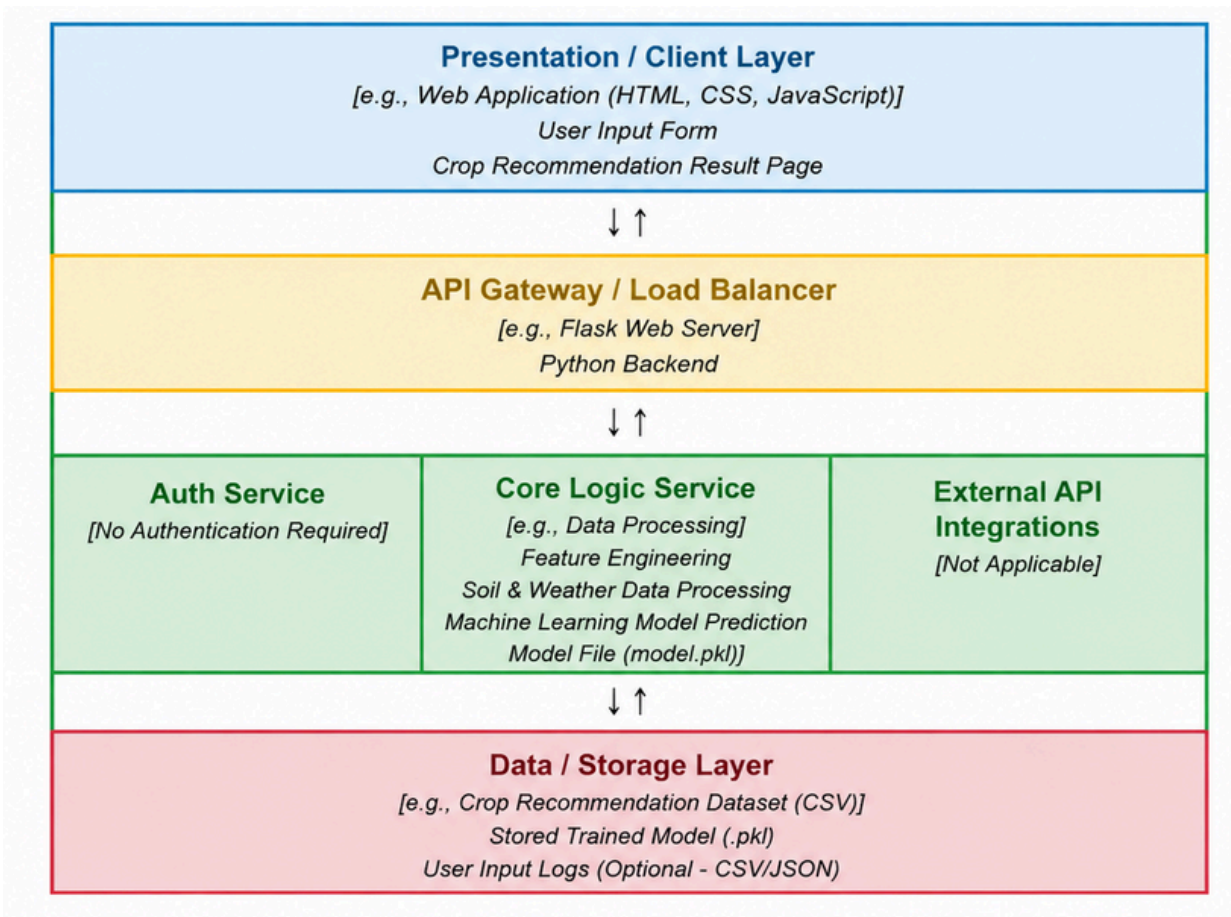


Solution Architecture

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides interface for user to enter soil and environmental parameters and view crop recommendation result.	HTML, CSS, JavaScript
API Gateway / Load Balancer	Receives user input from the web interface and sends data to the backend for crop prediction.	Flask, Python
Microservices (Application Layer)	Processes input data, performs feature engineering, and uses machine learning model to predict the most suitable crop.	Scikit-learn, Pandas, NumPy
Data / Storage Layer	Stores crop dataset, trained machine learning model, and optional user input logs.	CSV Files, Pickle Model (.pkl)